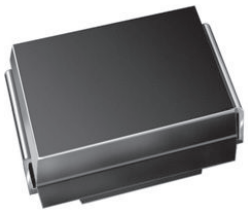
Surface-Mount Ultrafast Plastic Rectifier



SMB (DO-214AA)

Cathode ○—▶|—○ Anode

LINKS TO ADDITIONAL RESOURCES

3D Models

FEATURES

- Glass passivated pellet chip junction
- Ideal for automated placement
- Ultrafast reverse recovery time
- Low switching losses, high efficiency
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

TYPICAL APPLICATIONS

For use in high frequency rectification and freewheeling application in switching mode converters and inverters for consumer, computer and telecommunication.

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	3.0 A
V_{RRM}	400 V, 600 V
I_{FSM}	35 A
t_{rr}	50 ns
V_F at I_F = 3.0 A	1.20 V
T_J max.	175 °C
Package	SMB (DO-214AA)
Circuit configuration	Single

MECHANICAL DATA

Case: SMB (DO-214AA)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes cathode end

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	MURS340S	MURS360S	UNIT
Device marking codes			3GS	3JS	
Maximum repetitive peak reverse voltage		V_{RRM}	400	600	V
Maximum average forward rectified current	T_M = 130 °C	$I_{F(AV)}$ (1)	3.0		A
	T_A = 25 °C	$I_{F(AV)}$ (2)	1.5		
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load		I_{FSM}	35		A
Operating junction and storage temperature range		T_J, T_{STG}	-65 to +175		°C

Notes

(1) Units mounted on PCB with 8 mm x 8 mm, 1 oz. copper pad areas (fig. 1)
(2) Free air, mounted on recommended copper pad area (fig. 2)

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**MURS340S**	**MURS360S**	**UNIT**
Maximum instantaneous forward voltage	I_F = 3.0 A	T_J = 25 °C	V_F (1)	1.45		V
		T_J = 150 °C		1.20		
Maximum instantaneous reverse current	Rated V_R	T_J = 25 °C	I_R (2)	5.0		μA
		T_J = 150 °C		150		
Maximum reverse recovery time	I_F = 0.5 A, I_R = 1.0 A, I_{rr} = 0.25 A		t_{rr}	50		ns
Maximum reverse recovery time	I_F = 1.0 A, dI/dt = 50 A/μs, V_R = 30 V, I_{rr} = 10 % I_{RM}		t_{rr}	75		ns

Notes

(1) Pulse test: 300 μs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)				
PARAMETER	**SYMBOL**	**MURS340S**	**MURS360S**	**UNIT**
Typical thermal resistance	$R_{\theta JM}$ (1)	12		°C/W
	$R_{\theta JA}$ (2)	120		

Notes

(1) Units mounted on PCB with 8 mm x 8 mm, 1 oz. copper pad areas. Thermal resistance $R_{\theta JM}$ - junction to mount

(2) Free air, mounted on recommended copper pad area. Thermal resistance $R_{\theta JA}$ - junction to ambient

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
MURS360S-M3/52T	0.093	52T	750	7" diameter plastic tape and reel
MURS360S-M3/5BT	0.093	5BT	3200	13" diameter plastic tape and reel

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

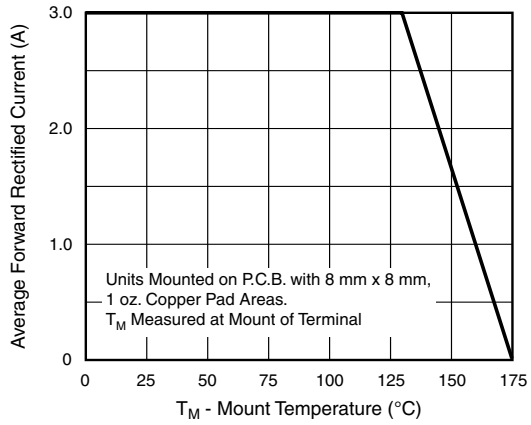


Fig. 1 - Forward Current Derating Curve

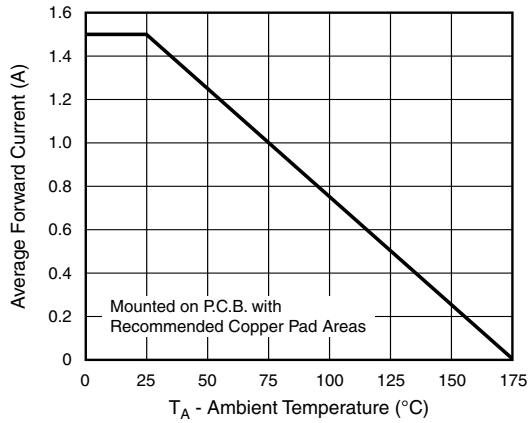


Fig. 2 - Forward Current Derating Curve

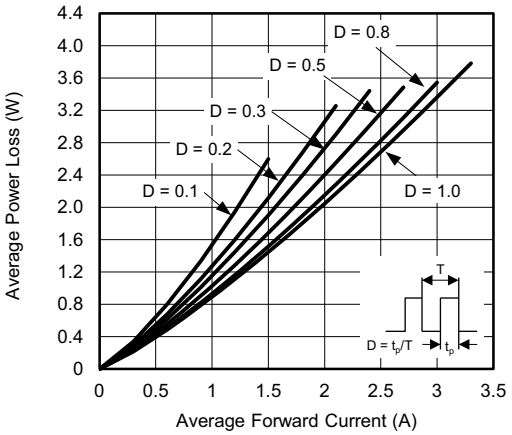


Fig. 3 - Forward Power Loss Characteristics

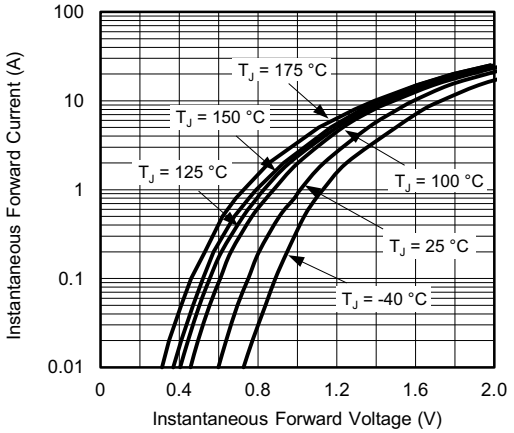


Fig. 4 - Typical Instantaneous Forward Characteristics

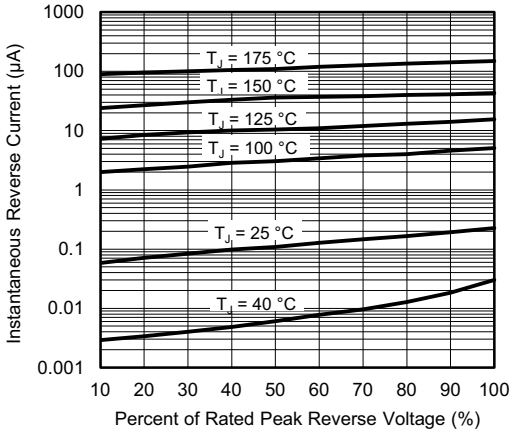


Fig. 5 - Typical Reverse Characteristics

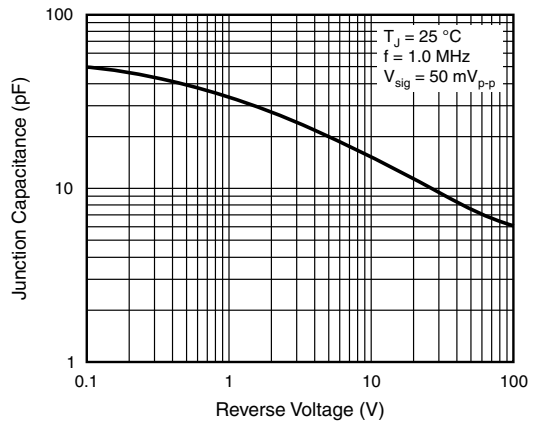
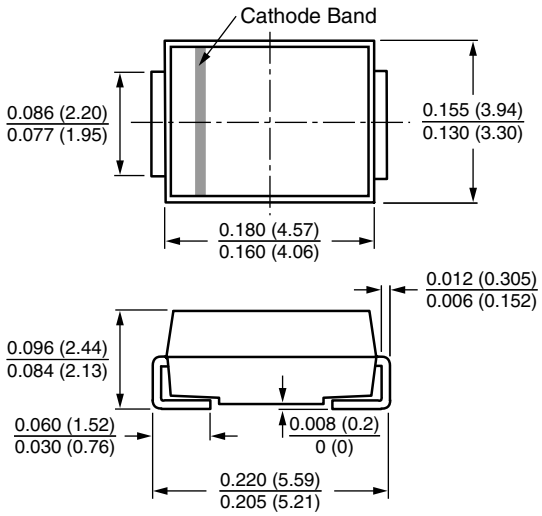


Fig. 6 - Typical Junction Capacitance

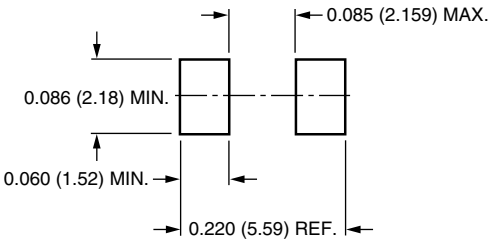
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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMB (DO-214AA)



Mounting Pad Layout



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